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Summary

- 📌 Robotician and networked control systems researcher specializing in edge/cloud-enabled autonomous systems, autonomous mobile robots, artificial intelligence, and communication-control-computation co-design. Research experience spans scalable and resilient robotic architectures, dynamic resource allocation, and industrial autonomous systems, with contributions across Swedish and EU-funded research projects, academic publications, and industrial collaborations.

Professional Experience

- Present
- 📌 **Experienced Researcher at Ericsson Research**, Cyber Physical Systems, Decision & Control, Stockholm, Sweden
 - *Development of concepts for solving challenges in communication-control-computation co-design*
 - *Evaluation and demonstration of solutions and algorithms in testbed environments*
 - *Publishing research through patents, top-tier conferences and journals*
 - *Supervision of M.Sc. students and Ph.D. interns*
 - *Engagement in internal and external research collaborations for driving the future of connected devices*
- 2025
- 📌 **CTO | Co-Founder at Eighth Dimension GmbH**, Munich, Germany | *Robotics startup*
 - *Module integration for detection and tracking of objects in real time*
 - *Module integration for multi-modal and multi-session 3D SLAM*
 - *Development of navigation and planning tools utilizing continuously updated 3D world models*
 - 📌 **Visiting Researcher at Epiroc**, Automation Lab, Örebro, Sweden
 - *Facilitate research directions on dynamic resource allocation for edge-enabled autonomous machines*
- 2023
- 📌 **Visiting Researcher at Ericsson Research**, Cloud and Technology, Lund, Sweden
 - *Development of Kubernetes-based scalable cloud-robot architectures for multi-agent systems*
- 2021
- 📌 **Automation Engineer at Rockwell Automation**, (I.C. SYSTEMS), Athens, Greece
 - *Tasty Foods S.A.: Development of SCADA systems*
 - *V. Bougatsos S.A.: Design and implementation of electrical panels | Development of SCADA systems*
 - *Motor Oil (Hellas), Corinth Refineries S.A.: Development of PLC systems | Design of electrical panels*

Education

- 2025
- 📌 **Ph.D.** at Luleå University of Technology, Robotics and AI research group, Luleå, Sweden
Thesis title: *Towards the Next Generation Robotic Autonomy on the Edge: Scalability, Resiliency, and Dynamic Resource Allocation*
- 2023
- 📌 **Licentiate** at Luleå University of Technology, Robotics and AI research group, Luleå, Sweden
Thesis title: *Towards Enabling the Next Generation of Edge Controlled Robotic Systems*
- 2020
- 📌 **Integrated M.Sc.** at University of Patras, Electrical and Computer Engineering, Patras, Greece
Thesis title: *Development and Control of an Advanced Robotic Tool for Enhanced Minimally Invasive Robotic Surgery*

Skills

Languages	📖 English Greek
Coding	📖 Python MATLAB Simulink C C++
Robotics	📖 ROS ROS2 CAD
Networks	📖 5G MQTT TCP/IP
Cloud/Edge	📖 Docker Kubernetes Helm
Industry	📖 PLC SCADA HMI Electrical Design
Misc.	📖 Academic Research Teaching Supervising Project Management

Projects

- 2024
- 📖 **SUNRISE-6G**, EU's Horizon Research and Innovation Programme, Ph.D. Researcher (1 year)
Project: *Edge-offloading and data processing for enhancing autonomous multi-agent robotic missions*
Team coordinator for the Robotics and AI group
 - 📖 **AMBITIOUS**, EU's Horizon Research and Innovation Programme, Ph.D. Researcher (1 year)
Project: *Autonomous UAV missions for IoT sensor data collection and operation*
Team coordinator for the Robotics and AI group
 - 📖 **PERSEPHONE**, EU's Horizon Research and Innovation Programme, Ph.D. Researcher (1 year)
Project: *Server-robot communication for enhanced autonomous robotic mining-drilling missions*
 - 📖 **ARCTIC EDGE**, EU's Northern Periphery and Arctic Programme, Ph.D. Researcher (1 year)
Project: *Implementation of a cloud-based, IoT, and MQTT technology-driven solution for rural manufacturing NPA SMEs, aiming to optimize supply chain processes*
 - 📖 **LIFELONG LEARNING**, Luleå University of Technology, Course Coordinator and Instructor (6 months)
Course: *Edge Computing in Robotics*
Course: *6G in Automation*
- 2023
- 📖 **REBOOT SKILLS**, Digital Europe Programme, Course Coordinator and Instructor (1 year)
Course: *Industrial Automation*
- 2021
- 📖 **AERO-TRAIN**, Marie-Sklodowska-Curie ITN - ETN project, Early-Stage Researcher (3 year)
Project: *Augmented reality for enhancing semi-autonomous remote aerial manipulation*
 - 📖 **5G EDGE INNOVATIONS FOR MINING**, Vinovva project, Ph.D. Researcher (3 year)
Project: *Development of 5G / edge-based architectures for enabling autonomous drone operations in underground mines*
 - 📖 **G-DRONES**, Vinovva project, Ph.D. Researcher (1.5 year)
Project: *Integrate seamless communication between a base station (Grafana interface) and drones through MQTT protocol for safe aerial inspection of open-pit mines*

Research Publications

Journal Articles

- 1 V. N. Sankaranarayanan, **A. S. Seisa**, A. Saradagi, S. G. Satpute, and G. Nikolakopoulos, "Towards Safety and Scalability in the Coordination of Coupled Aerial-Ground Robots," (*submitted*), 2026.
- 2 **A. S. Seisa**, S. Velhal, S. Kotpalliwar, S. Satpute, and G. Nikolakopoulos, "Optimization of Edge-Offloading for Centralized Controllers Through Dynamic Computational Resource Allocation," *IEEE Internet of Things Journal*, 2026. 📄 DOI: 10.1109/JIOT.2026.3650978.
- 3 A. Berra, J. Mellet, **A. S. Seisa**, *et al.*, "Towards Advanced Aerial Physical Inspection in Real Industrial Conditions," (*submitted*), 2025.

- 4 M.-N. Stamatopoulos, P. Koustoumpardis, **A. Seisa**, and G. Nikolakopoulos, "Path Planning for Collaborative System of Tethered UAVs in Dynamic and Confined Environments," (*submitted*), 2025.
- 5 **A. S. Seisa**, B. Lindqvist, S. G. Satpute, and G. Nikolakopoulos, "An Edge Architecture for Enabling Autonomous Aerial Navigation with Embedded Collision Avoidance Through Remote Nonlinear Model Predictive Control," *Journal of Parallel and Distributed Computing*, p. 104 849, 2024.  DOI: 10.1016/j.jpdc.2024.104849.
- 6 **A. S. Seisa**, B. Lindqvist, S. G. Satpute, and G. Nikolakopoulos, "E-CNMPC: Edge-Based Centralized Nonlinear Model Predictive Control for Multi-Agent Robotic Systems," *IEEE Access*, vol. 10, pp. 121 590–121 601, 2022.  DOI: 10.1109/ACCESS.2022.3223446.
- 7 **A. S. Seisa**, S. G. Satpute, B. Lindqvist, and G. Nikolakopoulos, "An Edge-Based Architecture for Offloading Model Predictive Control for UAVs," *Robotics*, vol. 11, no. 4, p. 80, 2022.  DOI: 10.3390/robotics11040080.

Conference Proceedings

- 1 G. Damigos, **A. S. Seisa**, N. Stathouloupoulos, S. Sandberg, and G. Nikolakopoulos, "Real-time Point Cloud Data Transmission via L4S for 5G-Edge-Assisted Robotics," in *IFAC World Congress*, 2026.
- 2 K. Nevala, S. B. Shahid, M. Tavangar, *et al.*, "Performance Improvements of 5G mmW Network with RIS in Underground Mine Environment," in *2026 Joint European Conference on Networks and Communications & 6G Summit (EuCNC/6G Summit)*, IEEE, 2026, pp. 1004–1009.  DOI: 10.1109/EuCNC/6GSummit68295.2026.11577357.
- 3 **A. S. Seisa**, E. Pagliari, G. Damigos, E. Small, and G. Nikolakopoulos, "Real-World Deployment of a 5G-Connected Edge-Controlled Aerial Robot in Industrial Subterranean Mines," in *2026 34th Mediterranean Conference on Control and Automation (MED)*, IEEE, 2026, pp. 521–526.  DOI: 10.1109/MED70602.2026.11598005.
- 4 **A. S. Seisa**, S. Kotpalliwar, S. G. Satpute, and G. Nikolakopoulos, "Dynamic Computational Resource Allocation for Ensuring Stability of Remote Edge-based Controlled Multi-agent Systems," in *2025 European Control Conference (ECC)*, IEEE, 2025, pp. 2907–2913.  DOI: 10.23919/ECC65951.2025.11187111.
- 5 **A. S. Seisa**, V. N. Sankaranarayanan, G. Damigos, S. G. Satpute, and G. Nikolakopoulos, "Cloud-Assisted Remote Control for Aerial Robots: From Theory to Proof-of-Concept Implementation," in *2025 IEEE 25th International Symposium on Cluster, Cloud and Internet Computing Workshops (CCGridW)*, 2025, pp. 171–176.  DOI: 10.1109/CCGridW65158.2025.00032.
- 6 A. Berra, V. N. Sankaranarayanan, **A. S. Seisa**, *et al.*, "Assisted Physical Interaction: Autonomous Aerial Robots with Neural Network Detection, Navigation, and Safety Layers," in *2024 International Conference on Unmanned Aircraft Systems (ICUAS)*, IEEE, 2024, pp. 1354–1361.  DOI: 10.1109/ICUAS60882.2024.10557050.
- 7 J. Mellet, A. Berra, **A. S. Seisa**, *et al.*, "Design of a Flexible Robot Arm for Safe Aerial Physical Interaction," in *2024 IEEE 7th International Conference on Soft Robotics (RoboSoft)*, IEEE, 2024, pp. 1048–1053.  DOI: 10.1109/RoboSoft60065.2024.10522019.
- 8 **A. S. Seisa**, S. G. Satpute, and G. Nikolakopoulos, "Cloud-Based Scheduling Mechanism for Scalable and Resource-Efficient Centralized Controllers," in *IECON 2024 - 50th Annual Conference of the IEEE Industrial Electronics Society*, 2024, pp. 1–6.  DOI: 10.1109/IECON55916.2024.10905254.
- 9 G. Damigos, **A. S. Seisa**, S. G. Satpute, T. Lindgren, and G. Nikolakopoulos, "A Resilient Framework for 5G-Edge-Connected UAVs Based on Switching Edge-MPC and Onboard-PID Control," in *2023 IEEE 32nd International Symposium on Industrial Electronics (ISIE)*, IEEE, 2023, pp. 1–8.  DOI: 10.1109/ISIE51358.2023.10228114.

- 10 V. N. Sankaranarayanan, G. Damigos, **A. S. Seisa**, S. G. Satpute, T. Lindgren, and G. Nikolakopoulos, "PACED-5G: Predictive Autonomous Control Using Edge for Drones over 5G," in *2023 International Conference on Unmanned Aircraft Systems (ICUAS)*, IEEE, 2023, pp. 1155–1161.  DOI: 10.1109/ICUAS57906.2023.10156241.
- 11 **A. S. Seisa**, N. Evangeliou, A. Tzes, and G. Nikolakopoulos, "Development and Experimental Evaluation of a 3DoF Tendon-Driven Probe for Robot Assisted Minimally Invasive Surgical Operations," in *2023 International Conference on Control, Automation and Diagnosis (ICCAD)*, IEEE, 2023, pp. 1–6.  DOI: 10.1109/ICCAD57653.2023.10152334.
- 12 M.-N. Stamatopoulos, P. Koustoumpardis, **A. Seisa**, and G. Nikolakopoulos, "Combined Aerial Cooperative Tethered Carrying and Path Planning for Quadrotors in Confined Environments," in *2023 31st Mediterranean Conference on Control and Automation (MED)*, 2023, pp. 364–369.  DOI: 10.1109/MED59994.2023.10185884.
- 13 **A. S. Seisa**, G. Damigos, S. G. Satpute, A. Koval, and G. Nikolakopoulos, "Edge Computing Architectures for Enabling the Realisation of the Next Generation Robotic Systems," in *2022 30th Mediterranean Conference on Control and Automation (MED)*, IEEE, 2022, pp. 487–493.  DOI: 10.1109/MED54222.2022.9837289.
- 14 **A. S. Seisa**, S. G. Satpute, B. Lindqvist, and G. Nikolakopoulos, "An Edge Architecture Oriented Model Predictive Control Scheme for an Autonomous UAV Mission," in *2022 IEEE 31st International Symposium on Industrial Electronics (ISIE)*, IEEE, 2022, pp. 1195–1201.  DOI: 10.1109/ISIE51582.2022.9831701.
- 15 **A. S. Seisa**, S. G. Satpute, and G. Nikolakopoulos, "A Kubernetes-Based Edge Architecture for Controlling the Trajectory of a Resource-Constrained Aerial Robot by Enabling Model Predictive Control," in *2022 26th International Conference on Circuits, Systems, Communications and Computers (CSCC)*, IEEE, 2022, pp. 290–295.  DOI: 10.1109/CSCC55931.2022.00056.
- 16 **A. S. Seisa**, S. G. Satpute, and G. Nikolakopoulos, "Comparison Between Docker and Kubernetes Based Edge Architectures for Enabling Remote Model Predictive Control for Aerial Robots," in *IECON 2022–48th Annual Conference of the IEEE Industrial Electronics Society*, IEEE, 2022, pp. 1–6.  DOI: 10.1109/IECON49645.2022.9968933.

Theses & Book Chapters

- 1 **A. S. Seisa**, "Towards the Next Generation Robotic Autonomy on the Edge: Scalability, Resiliency, and Dynamic Resource Allocation," in Luleå University of Technology, 2025.
- 2 **A. S. Seisa**, "Towards enabling the next generation of edge controlled robotic systems," in Luleå University of Technology, 2023.
- 3 **A. S. Seisa** and A. Koval, "Edge Connected ARWs," in *Aerial Robotic Workers*, Elsevier, 2023, pp. 245–253.  DOI: 10.1016/B978-0-12-814909-6.00019-6.
- 4 **A. S. Seisa**, "Development and control of an advanced robotic tool for enhanced minimally invasive robotic surgery," in University of Patras, 2020.

Academic & Professional Activities

Honors & Awards

- 2024  **IVA's 100 List** Reconfigurable 6G Quality of Service (QoS) and Edge Integration for Robotics and Automation

Academic & Professional Activities (continued)

Lecturing & Teaching

- 2025  **AI in the Process Industry and Automation** (R7022E), Luleå University of Technology
- 2025  **Robotics for all course** (R0006E), Luleå University of Technology
- 2025  **Industrial Automation course** (R7008E), Luleå University of Technology
- 2024  **Industrial Automation course** (R7008E), Luleå University of Technology
- 2024  **Robotics for all course** (R0006E), Luleå University of Technology
- 2023  **Industrial Automation course** (R7008E), Luleå University of Technology
- 2023  **Robotics for all course** (R0006E), Luleå University of Technology
- 2022  **Industrial Automation course** (R7008E), Luleå University of Technology

Talks

- 2025  **Enabling 5G/Edge Connected Robots within Subterranean Environments**, Keynote at the IEEE International Symposium on Cluster, Cloud, and Internet Computing (CCGrid 2025) RISE Workshop, Trømsø, Norway
- 2024  **Cloud-enabled Remote Control**, Presentation at the AERO-TRAIN Summer School 2024, Human-Robot Interaction Day, Crete, Greece
- 2023  **Edge Connected Drones**, Seminar at the Automatic Control Group 2023, Lund University, Lund, Sweden

Supervision

- 2025  **Endemann, Moritz**, Multiple Edge Robots for World Mapping
- 2023  **Markostamos, Georgios**, Autonomous Multi-Agent Exploration and Mapping
- 2022  **Stamatopoulos, Marios-Nektarios**, Collaborative Control of Aerial Robots (drones) for Co-Manipulating Flexible Objects (ropes)
- 2022  **Dahlquist, Niklas**, Generation of Behavior Trees for Dynamic Agents Based on Market Inspired Task Allocation

Workshops & Events

- 2025  **Full-Day CCGrid 2025 Workshop**, Committee at the “RISE” - Robotic Systems in the Edge-Cloud Continuum Workshop, Trømsø, Norway
- 2024  **Full-Day ICUAS 2024 Workshop**, Organizer at the Aerial Workers for Infrastructure and Asset Maintenance: The journey from “Lab” to “Real-World” Workshop, Crete, Greece
- 2024  **AERO-TRAIN Summer School 2024**, Presenter at the Aerial robots and their role in inspection and maintenance of industrial plants, Crete, Greece
- 2023  **Mohamed Bin Zayed International Robotics Challenge 2023** Team ROC